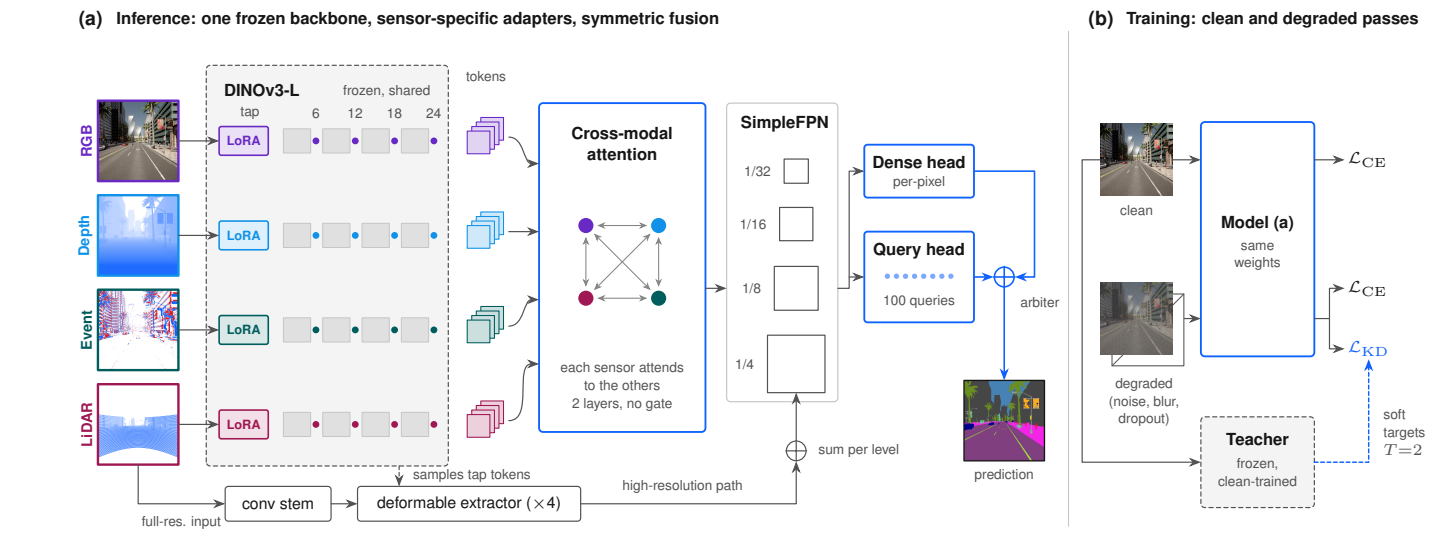

(a) Inference: one frozen backbone, sensor-specific adapters, symmetric fusion
DINOv3-L
frozen, shared
tap
6
12
18
24
RGB
Depth
Event
LiDAR
LoRA
LoRA
LoRA
LoRA
tokens
Cross-modal attention
each sensor attends to the others
2 layers, no gate
SimpleFPN
1/32
1/16
1/8
1/4
Dense head
per-pixel
Query head
100 queries
arbiter
prediction
sum per level
samples tap tokens
high-resolution path
full-res. input
conv stem
deformable extractor (×4)
(b) Training: clean and degraded passes
clean
degraded
(noise, blur, dropout)
Model (a)
same weights
$\mathcal{L}_{\mathrm{CE}}$
$\mathcal{L}_{\mathrm{CE}}$
$\mathcal{L}_{\mathrm{KD}}$
Teacher
frozen, clean-trained
soft targets $T=2$